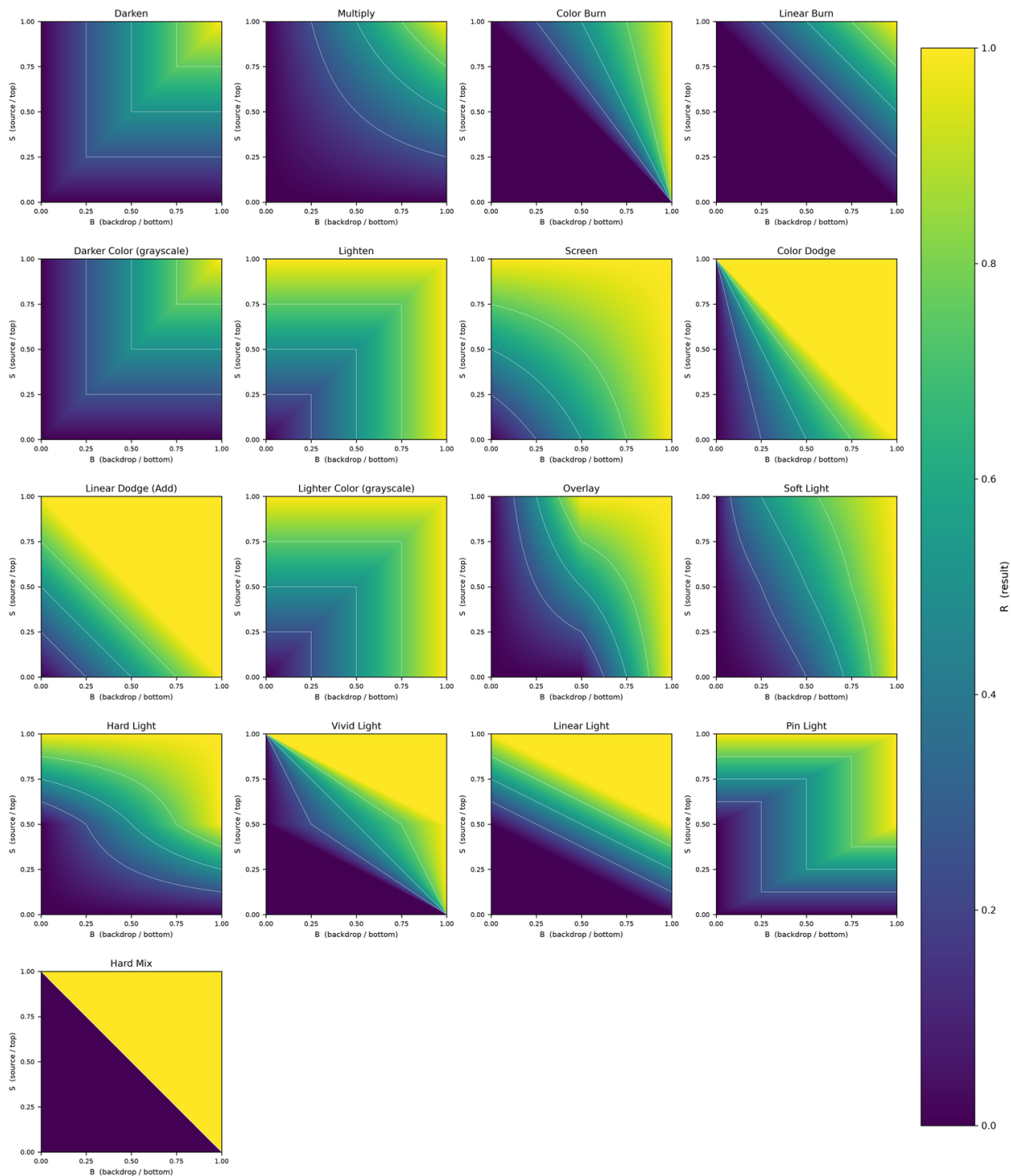


混合模式函数图谱

Blend Mode Function Atlas

中英文对照版 | Bilingual Chinese-English Edition

Blend mode functions — x: B, y: S, color: R



阅读方法 | How to Read the Maps

每张图都把底层（背景）值 B 放在横轴，把上层（源）值 S 放在纵轴。两者与输出 R 均已归一化到 $[0,1]$ 。颜色由深紫到黄色，对应 R 从 0 到 1；白色等值线标出 $R=.25$ 、 $.5$ 和 $.75$ 。

Each map places backdrop value B on the horizontal axis and source value S on the vertical axis. B , S , and result R are normalized to $[0,1]$. The shared color scale runs from dark purple at $R=0$ to yellow at $R=1$; white contours mark $R=.25$, $.5$, and $.75$.

边界与截断 | Boundaries and Clamping

除法类模式需要显式处理分母为零的边界。Color Burn 在 $S=0$ 时定义为 0；Color Dodge 在 $S=1$ 时定义为 1。Linear Burn、Linear Dodge、Linear Light 以及所有可能越界的中间结果均截断到 $[0,1]$ 。Soft Light 采用 PDF/W3C 的分段函数，其中 $D(B)=((16B-12)B+4)B$ ($B \leq .25$)，否则 $D(B)=\sqrt{B}$ 。

Division-based modes require explicit zero-denominator boundaries. Color Burn is defined as 0 at $S=0$; Color Dodge is defined as 1 at $S=1$. Linear Burn, Linear Dodge, Linear Light, and all potentially out-of-range intermediate results are clamped to $[0,1]$. Soft Light uses the PDF/W3C piecewise function, with $D(B)=((16B-12)B+4)B$ for $B \leq .25$ and $D(B)=\sqrt{B}$ otherwise.

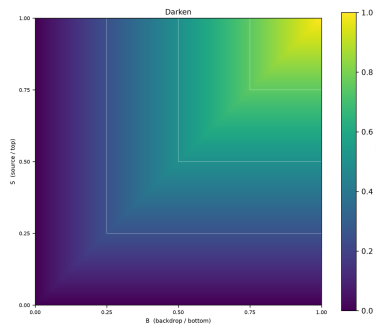
关于 Darker Color / Lighter Color | Whole-pixel Comparison

Darker Color 与 Lighter Color 在 RGB 图像中比较整像素的亮度并整体选择一个像素，因此无法由单通道二维函数完整表达。本图谱展示灰度情形；此时它们分别与 Darken、Lighten 相同。

In RGB images, Darker Color and Lighter Color compare whole-pixel luminosity and select one complete pixel. A scalar 2D function cannot fully represent that behavior. The atlas therefore shows the grayscale case, where they reduce to Darken and Lighten respectively.

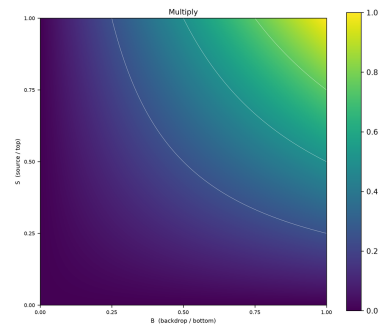
变暗组 | Darken Group

变暗 Darken

 $\min(B, S)$ 

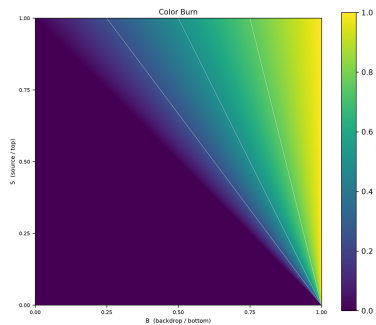
逐通道选择较小值。
Select the smaller value per channel.

正片叠底 Multiply

 $B * S$ 

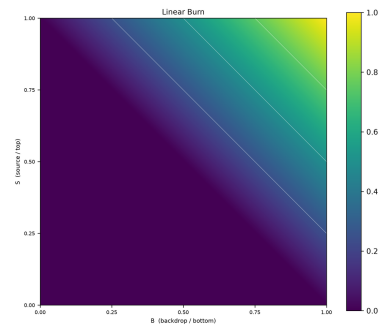
总是不亮于任一输入。
Never brighter than either input.

颜色加深 Color Burn

 $S=0: 0; \text{ else } \max(0, 1-(1-B)/S)$ 

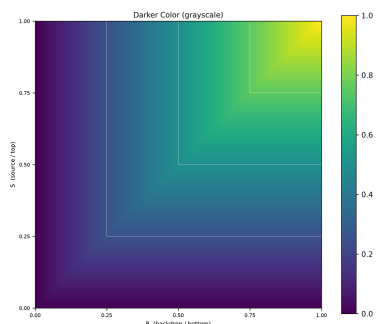
$S=0$ 单独定义为 0，并在下界截断。
 $S=0$ is defined as 0; clamp at the lower bound.

线性加深 Linear Burn

 $\text{clamp}(B+S-1)$ 

在线性相加后减 1，并截断到 [0,1]。
Add linearly, subtract 1, then clamp to [0,1].

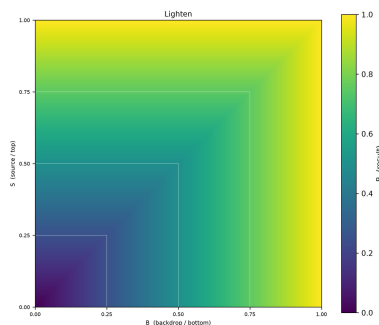
深色（灰度） Darker Color (grayscale)

 $\min(B, S)$ 

RGB 版本比较整像素；灰度图退化为 Darken。
The RGB mode compares whole pixels; grayscale reduces to Darken.

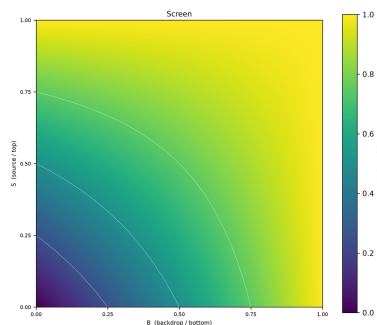
变亮组 | Lighten Group

变亮 Lighten

 $\max(B, S)$ 

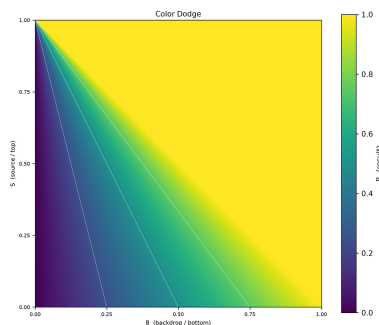
逐通道选择较大值。
Select the larger value per channel.

滤色 Screen

 $1 - (1 - B)(1 - S)$ 

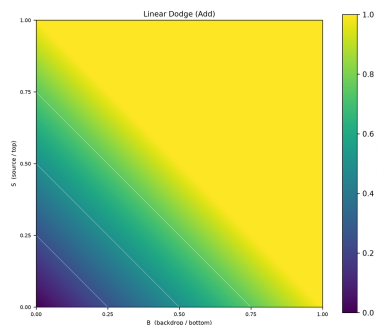
对补色相乘后再取补色。
Multiply complements, then take the complement.

颜色减淡 Color Dodge

 $S=1: 1; \text{ else } \min(1, B/(1-S))$ 

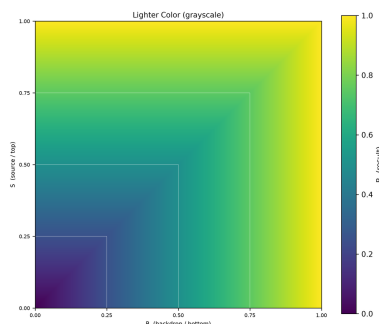
S=1 单独定义为 1，并在上界截断。
S=1 is defined as 1; clamp at the upper bound.

线性减淡（添加） Linear Dodge (Add)

 $\text{clamp}(B+S)$ 

直接相加，并截断到 [0,1]。
Add directly, then clamp to [0,1].

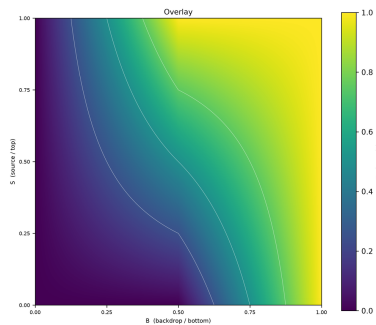
浅色（灰度） Lighter Color (grayscale)

 $\max(B, S)$ 

RGB 版本比较整像素；灰度图退化为 Lighten。
The RGB mode compares whole pixels; grayscale reduces to Lighten.

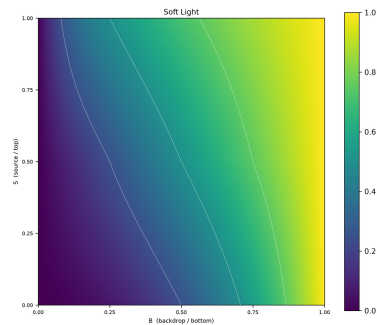
对比与光照组 I | Contrast and Light Group I

叠加 Overlay

$$B \leq .5: 2BS; \text{ else } 1 - 2(1-B)(1-S)$$


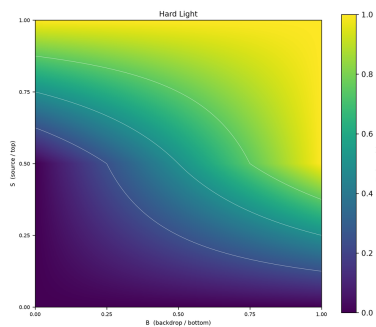
分支由底层 B 决定；B=.5 时结果等于 S。
The backdrop B selects the branch; at B=.5 the result equals S.

柔光 Soft Light

$$S \leq .5: B - (1-2S)B(1-B); \text{ else } B + (2S-1)(D(B)-B)$$


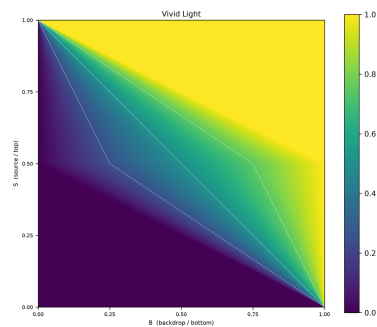
采用 PDF/W3C 分段定义；D(B) 还在 B=.25 处分段。
Uses the PDF/W3C piecewise definition; D(B) has another split at B=.25.

强光 Hard Light

$$S \leq .5: 2BS; \text{ else } 1 - 2(1-B)(1-S)$$


分支由上层 S 决定；等价于交换输入后的 Overlay。
The source S selects the branch; equivalent to Overlay with inputs exchanged.

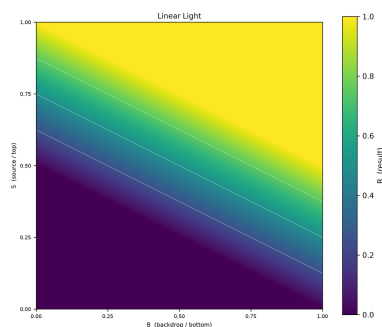
亮光 Vivid Light

$$S \leq .5: \text{Burn}(B, 2S); \text{ else } \text{Dodge}(B, 2S-1)$$


下半区使用颜色加深，上半区使用颜色减淡。
Uses Color Burn below the midpoint and Color Dodge above it.

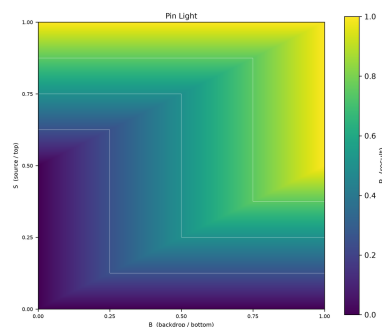
对比与光照组 II | Contrast and Light Group II

线性光 Linear Light

 $\text{clamp}(B+2S-1)$ 

以 $S=.5$ 为中性点的线性加减。
Linear addition/subtraction with $S=.5$ as the neutral point.

点光 Pin Light

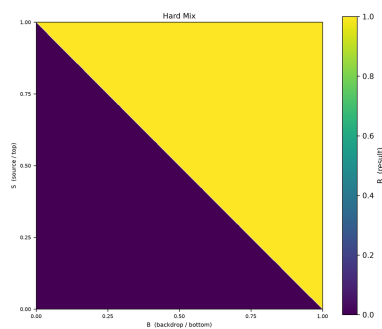
 $S < .5: \min(B, 2S); \text{ else } \max(B, 2S-1)$ 

在中点两侧分别以 min 和 max 替换像素。
Uses min below the midpoint and max above it.

阈值模式 | Threshold Mode

实色混合 Hard Mix

```
VividLight(B,S)<.5: 0; else 1
```



对亮光结果在 .5 处二值化；等于 .5 时取 1。
Threshold Vivid Light at .5; equality maps to 1.

校验摘要 | Verification Summary

函数使用 1001 x 1001 的规则网格采样。所有模式的输出均为有限值，范围均落在 [0,1]。自动断言覆盖了 Overlay 与 Hard Light 的分支方向和中点连续性、Soft Light 的两个 D(B) 区间、Color Burn / Dodge 的除零边界、Vivid Light 的端点与中性点、Pin Light 的上下分支，以及 Hard Mix 的 .5 阈值约定。

Functions were sampled on a regular 1001 x 1001 grid. Every mode produced finite values within [0,1]. Automated assertions cover Overlay and Hard Light branch orientation and midpoint continuity, both D(B) regions of Soft Light, the zero-denominator boundaries of Color Burn and Dodge, Vivid Light endpoints and neutral point, both Pin Light branches, and the .5 threshold convention of Hard Mix.

复现 | Reproduction

运行 `generate_blend_mode_maps.py` 可重新生成总览图、全部单图与文本校验报告。PDF 由 `build_bilingual_pdf.py` 从这些图像构建。

Run `generate_blend_mode_maps.py` to regenerate the overview, individual maps, and text verification report. The PDF is assembled from those figures by `build_bilingual_pdf.py`.